

D365 F&O Warehouse Management system (Advance warehouse)-2

This is the continuation of the previous article on warehouse management series.

Reservation Hierarchy

Reservation hierarchy models different levels on which reservations can be made. Each level represents a physical storage or tracking dimension in inventory. Reservation hierarchy is a required setup for warehouse enabled products.

The advanced warehousing reservation functionality allows items enabled for advanced warehousing (AWMS), the ability to use flexible warehouse operations, postponement of reservation of specific inventory dimensions until allocation is complete, and gives clear separation of which inventory dimensions can be specified, and when they can be specified.

Create A Reservation Hierarchy

To create a new reservation hierarchy,

Navigation: Warehouse management > Setup > Inventory > Reservation hierarchy.

Click the “New” button and give the reservation hierarchy an ID and meaningful description.

Any dimensions that you will not be using for your products can be moved over the “Available” pane to the left using the left arrow button un the middle of the form.

For example, if your products do not use serial numbers, vendor consignment, or batch number (lot numbers) you can select the “Batch number”, “Serial number”, and “Owner” dimensions and click the left arrow button. Then click the “Ok” button to create the reservation hierarchy.

New reservation hierarchy

Name: Description:

AVAILABLE


We didn't find anything to show here.

SELECTED

Site
Warehouse
Inventory status
Location
License plate
Batch number
Serial number
Owner

OK Cancel

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Inventory reservation hierarchies

Name: Batch Description: Batch reservation hierarchy

Hierarchy

Name	Reservation hierarchy level	Allow reservation on demand order
Site	1	☒
Warehouse	2	☒
Inventory status	3	☒
Location	4	☒
License plate	5	☒
Batch number	6	☒

The reservation hierarchy setup allows you to reserve inventory at different levels without providing details about where to reserve and what dimensions to reserve on. This lets you postpone specific details, such as the location or license plate to make the reservation on. The location to reserve items on can be determined later by the location directive at the time the order is released for picking.

For example, in the figure above, any dimension below the location dimension in the reservation hierarchy will not be specifically reserved until the order is allocated to the warehouse and warehouse work is generated for the warehouse worker to complete on the warehouse mobile application. Any dimension above the location dimension will be specifically reserved at the order level prior to warehouse release.

Allow reservation on demand order – This feature allows flexibility in the way that dimensions below the location dimension are reserved. For example, in the figure shown above, the batch dimension is below location and the “Allow reservation on demand order” is checked for the batch dimension. This means that on a sales order I can either reserve down to a specific batch or I can let the system reserve a batch based on location directive settings. This feature can allow you to get the best of both worlds so to speak. If you have FEFO requirements I would recommend setting the batch above location.

To fully understand what reservation hierarchy is and what it does, it is useful to see a few examples to see how they differ. We will look at two different configurations of the configuration hierarchy where the batch dimension is below location and above the location dimension.

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Reservation Hierarchy: Batch Below

In this example I have created a product and assigned it the reservation hierarchy “Batch”. This reservation hierarchy has the batch dimension set below the location dimension in the reservation hierarchy.

Released product details | My view

Z002 : Test Product 2

General

IDENTIFICATION

Product type
Item

Product subtype
Product

Product service type
Not specified

Item number
Z002

Product number
Z002

Engineering product category de...
Product owner

FURTHER IDENTIFICATION

Product name
Test Product 2

Search name
TestProduct2

Description

ADMINISTRATION

Storage dimension group
Ware

Tracking dimension group
Batch-Phy

Item model group
PWA

Reservation hierarchy
Batch

Counting reason code policy

Product lifecycle state

Advanced notes group

BUNDLE

Bundle
No

RELEASE

Release policy

As you can see the batch dimension is below the location dimension.

Save + New Delete Options

batch reser

Batch
Batch reservation hierarchy

Inventory reservation hierarchies | Standard view

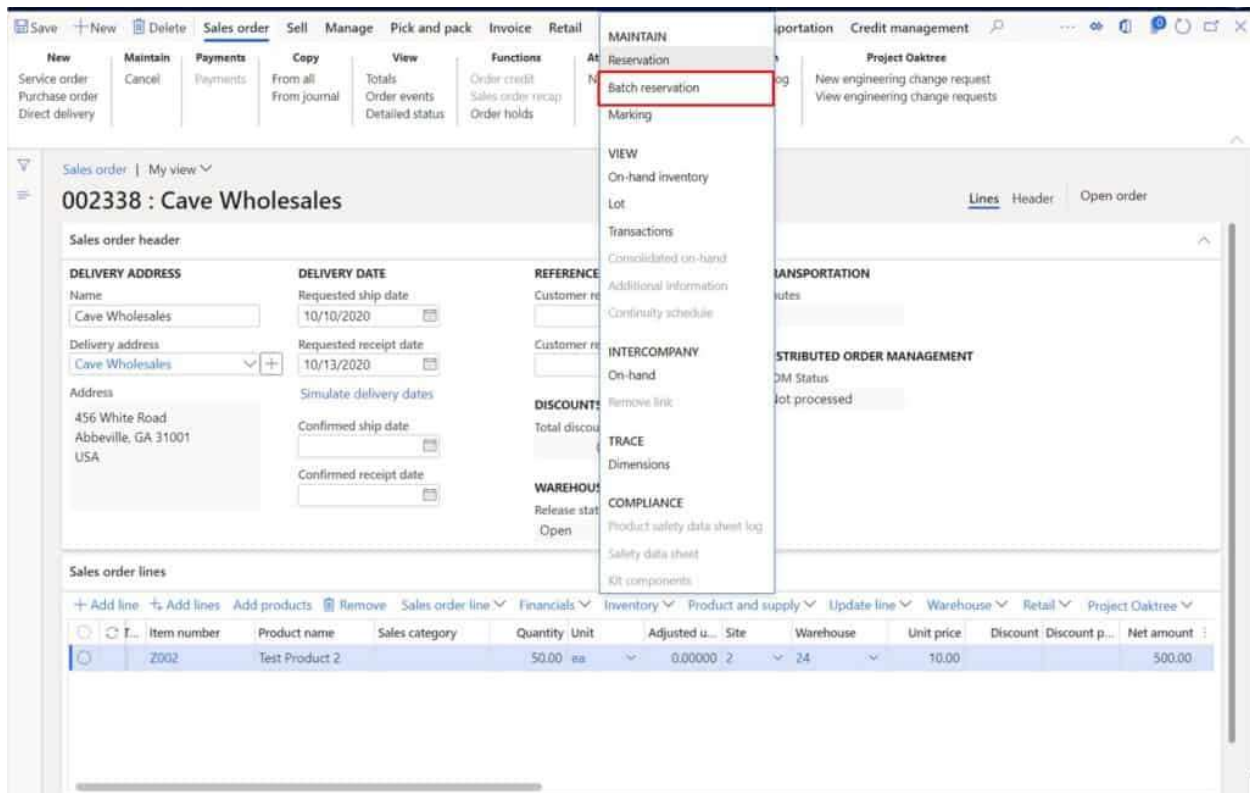
Name	Description
Batch	Batch reservation hierarchy

Hierarchy

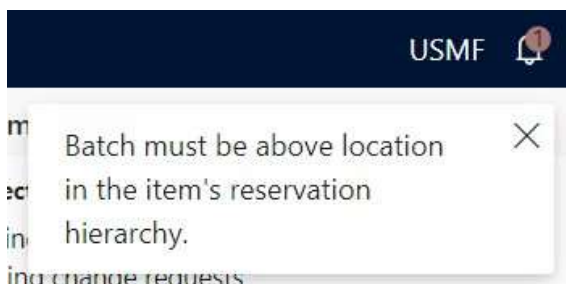
Name	Reservatio...	Allow reservation on demand order
Site	1	✓
Warehouse	2	✓
Inventory status	3	✓
Location	4	
License plate	5	
Batch number	6	

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Now I will add this item to a sales order and attempt to reserve to a specific batch by clicking the “Inventory” button on the sales line and clicking “Batch reservation” button.



Now I will receive an error telling me that I cannot reserve a specific batch because my batch dimension is below the location level.



I can reserve my sales order line by clicking the “Inventory” button on the sales line and clicking “Reservation” button.

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Sales order header

DELIVERY ADDRESS
 Name: Cave Wholesales
 Delivery address: Cave Wholesales
 Address: 456 White Road, Abbeville, GA 31001, USA

DELIVERY DATE
 Requested ship date: 10/10/2020
 Requested receipt date: 10/13/2020
 Confirmed ship date:
 Confirmed receipt date:

REFERENCE
 Customer re:
 Customer re:

DISCOUNTS
 Total discount:

WAREHOUSE
 Release status:
 Open:

Sales order lines

Item number	Product name	Sales category	Quantity	Unit	Adjusted u...	Site	Warehouse	Unit price	Discount	Discount p...	Net amount
2002	Test Product 2		50.00	ea	0.00000	2	24	10.00			500.00

Here I can reserve to the site, warehouse, and inventory status level by clicking “Reserve lot”.

Reserve lot | Options

002338 : 2002 | Standard view

Reservation

Unit symbol: ea
 Reserved physical: 50.00
 Reserved ordered: On order

Detailed availability by dimension

Refresh dimensions

☒ Site ☒ Warehouse ☒ Inventory status

Unit symbol	Item number	Maximum ...	Total avail...	Site	Warehouse	Inventory status	Reservation
ea	2002	950.00	950.00	2	24	Available	50.00

Source line dimensions
 Site=2, Warehouse=24, Inventory status=Available

Lock reservations

Quantity	Site	Warehouse	Inventory ...
50.00	☒	☒	☒

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Now when I release to warehouse, I get sales picking work but when I look at the batch dimension it is blank. That is because with batch below location, I did not reserve a specific batch, and I can simply pick the first batch I find.

Note: If you have FEFO (First Expiry First Out) requirements you will want the batch to be above location so that you can reserve a specific batch based on the expiration date.

The screenshot shows the 'Work' section of the D365 F&O Warehouse Management system. The 'Overview' tab is active, displaying a table with work order details. Below this, the 'Lines' tab is active, showing a table of work lines. The 'Batch number' column in the 'Lines' table is highlighted with a red box.

Work prior...	Locked by	Work ID	Work status	Site	Warehouse	Work pool ID	Target license plate...	Work order type	Order number
50		USMF-002342	Open	2	24			Sales orders	002338

Work status	Work type	Location	Item number	Product name	Work quan...	Remaining...	Unit	Batch number	User ID	Stop t
Open	Pick	BULK-005	Z002	Test Product 2	50.00	50.00	ea			
Open	Print		Z002	Test Product 2	50.00	50.00	ea			
Open	Put	BAYDOOR	Z002	Test Product 2	50.00	50.00	ea			

Key points to consider.

System will automatically take the first expired batch at backend if we have below setup at item model group.

RESERVATION

- ☐ FIFO date-controlled
- ☐ Backward from ship date

Item sales reservation

Default

- ☐ Same batch selection
- ☐ Consolidate requirement

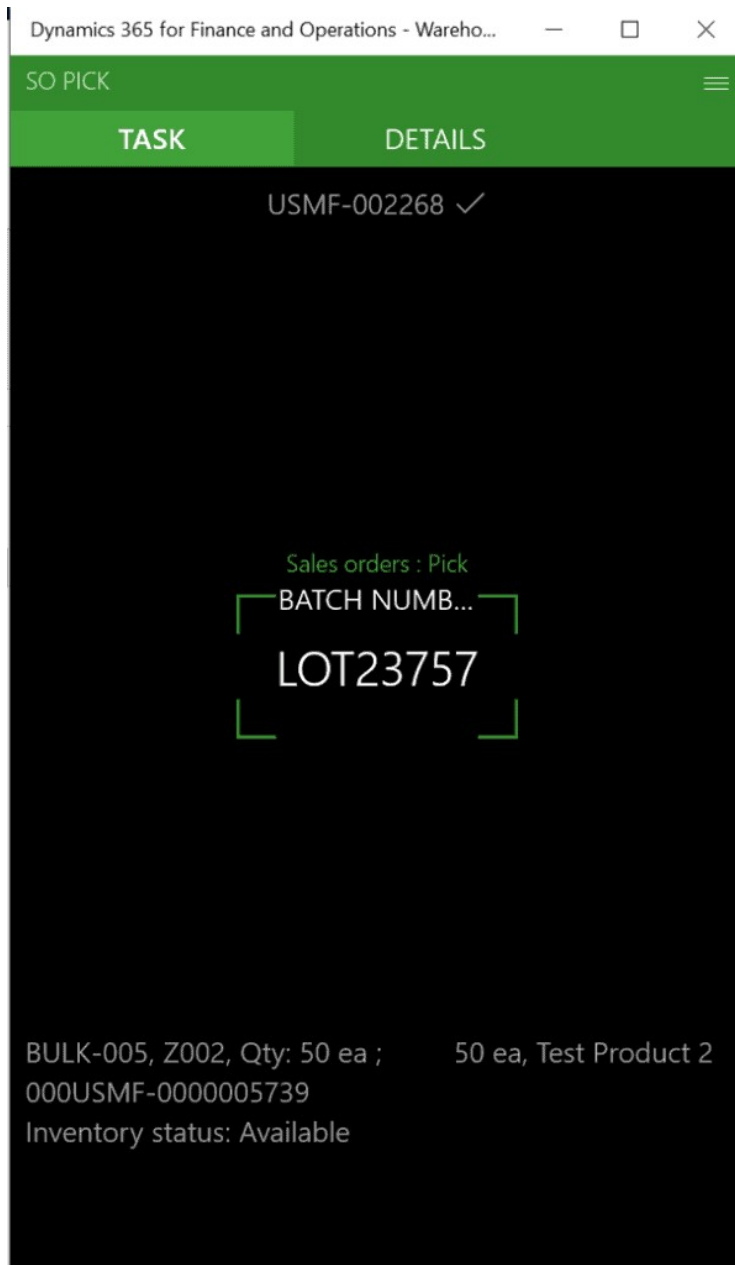
- ☒ FEFO date-controlled

Pick criteria

Expiration date

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Now when complete the work on the Dynamics 365 Finance and Operations warehouse mobile application I am prompted to select the batch I want to pick. So flexibility to select the batch is available.



Reservation Hierarchy: Batch Above

In this example I have created a product and assigned it the reservation hierarchy "BatchAbove". This reservation hierarchy has the batch dimension set above the location dimension in the reservation hierarchy.

D365 F&O Warehouse Management system (Advance warehouse)-2

Released product details | My view

Z001 : Test Product 1

General

IDENTIFICATION

Product type: Item
Product subtype: Product
Product service type: Not specified
Item number: Z001
Product number: Z001

Engineering product category de...
Product owner: Test Product 1
Search name: TestProduct1
Description: TestProduct1

ADMINISTRATION

Storage dimension group: Ware
Tracking dimension group: Batch-Phy
Item model group: PWA
Reservation hierarchy: BatchAbove
Counting reason code policy

Product lifecycle state:
Advanced notes group:
BUNDLE
Bundle: No
RELEASE
Release policy:

PURCHASE

PURCHASE ORDER
Unit: ea
Overdeliverv:

Item price tolerance group:
Vendor:
Price: 0.00
Price quantity:
Line discount group:
Multiline discount:

As you can see the batch dimension is above the location dimension.

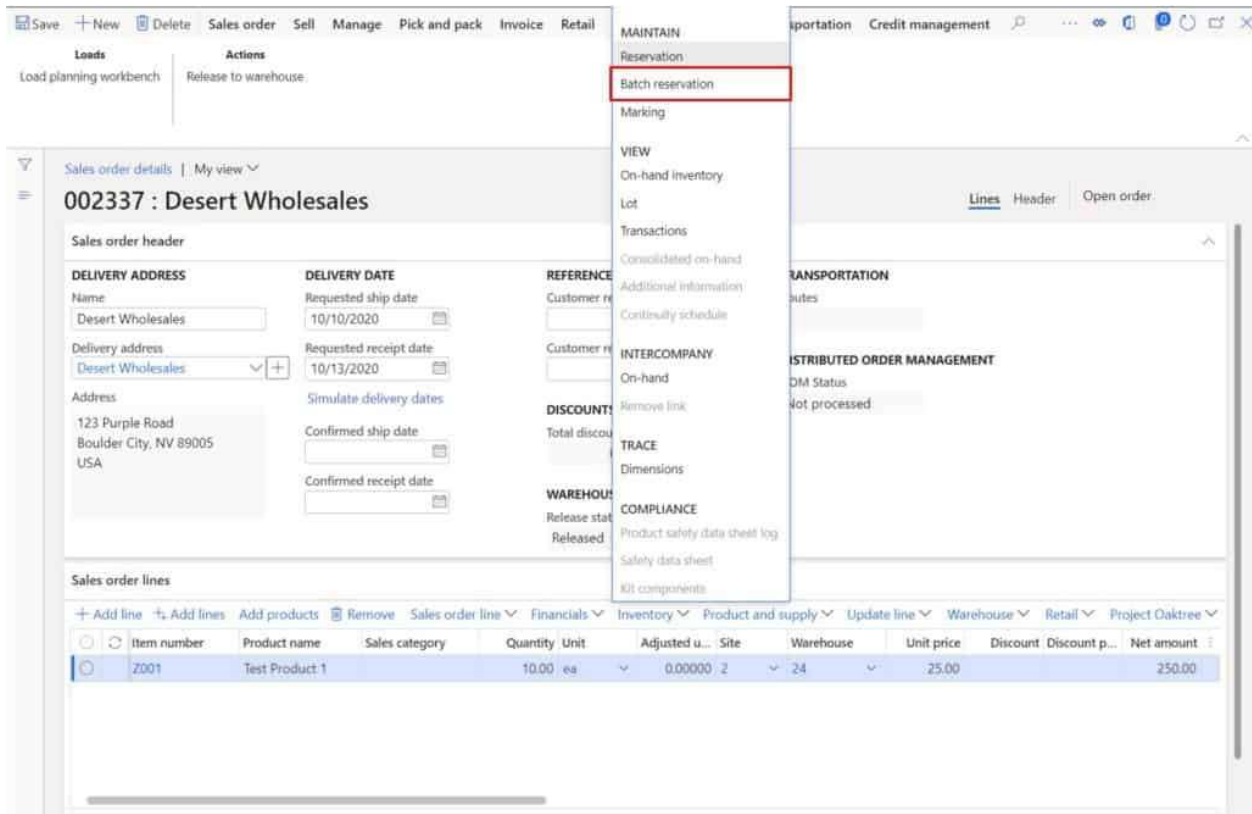
Inventory reservation hierarchies | Standard view

Name: BatchAbove Description: Batch above hierarchy

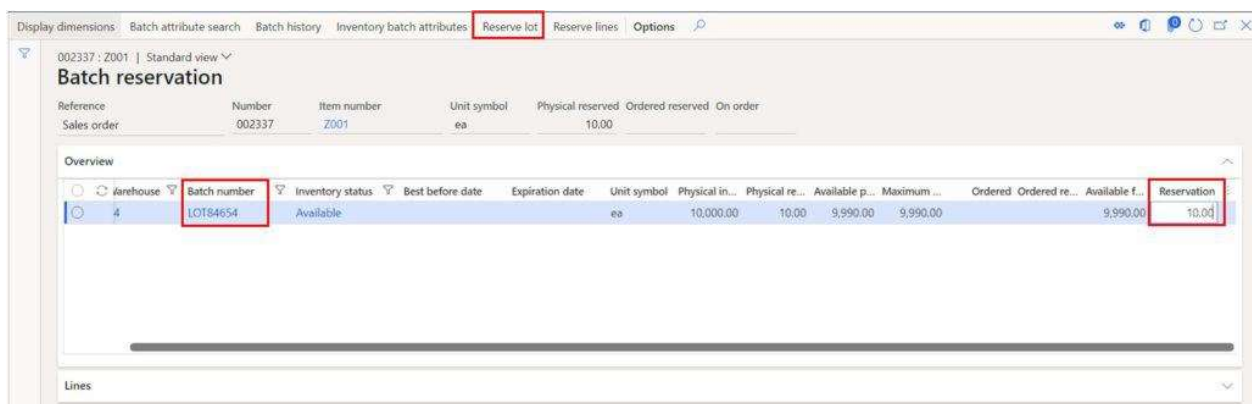
Name	Reservatio...	Allow reservation on demand order
Site	1	✓
Warehouse	2	✓
Inventory status	3	✓
Batch number	4	✓
Location	5	✓
License plate	6	

Now I will add this item to a sales order and attempt to reserve to a specific batch by clicking the "Inventory" button on the sales line and clicking "Batch reservation" button.

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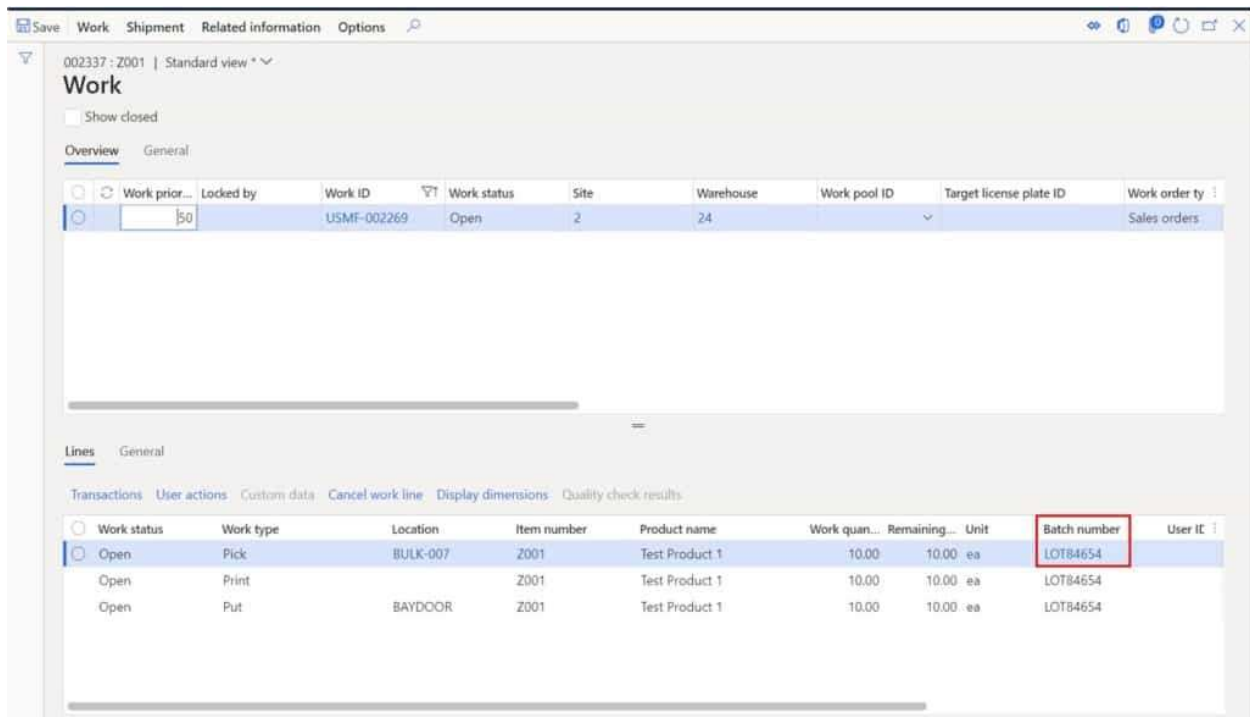


Now that I have batch above the location dimension, I am able to open the batch reservation form where I can see individual batches that are available for reservation. You can click the reserve lot button to reserve based on FEFO if you have enabled that in the item model group, or you can simply enter the quantity you want to reserve against the batch in the "Reservation" field. You can also reserve over multiple batches by entering the partial quantities against each batch that you want to reserve.



Now when I release this order to the warehouse, I get picking work that specifies the batch on the pick lines.

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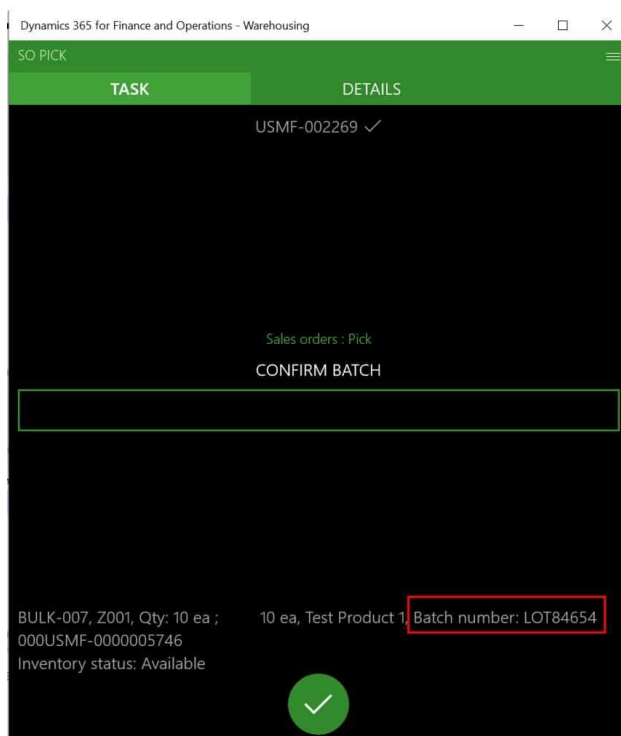


The screenshot shows the Dynamics 365 F&O Warehouse Management system interface. The top navigation bar includes 'Save', 'Work', 'Shipment', 'Related information', and 'Options'. The main area is titled '002337 : Z001 | Standard view' and has a 'Work' tab selected. Below the 'Work' tab, there is a 'Show closed' checkbox and an 'Overview' section. The 'Overview' section contains a table with columns: 'Work prior...', 'Locked by', 'Work ID', 'Work status', 'Site', 'Warehouse', 'Work pool ID', 'Target license plate ID', and 'Work order ty...'. The table has one row with values: '50', 'USMF-002269', 'Open', '2', '24', and 'Sales orders'. Below the 'Overview' section, there is a 'Lines' tab selected. The 'Lines' tab has a sub-section 'General' and a table with columns: 'Work status', 'Work type', 'Location', 'Item number', 'Product name', 'Work quan...', 'Remaining...', 'Unit', 'Batch number', and 'User ID'. The table has three rows: 'Open', 'Pick', 'BULK-007', 'Z001', 'Test Product 1', '10.00', '10.00', 'ea', 'LOT84654'; 'Open', 'Print', 'Z001', 'Test Product 1', '10.00', '10.00', 'ea', 'LOT84654'; and 'Open', 'Put', 'BAYDOOR', 'Z001', 'Test Product 1', '10.00', '10.00', 'ea', 'LOT84654'. The 'Batch number' column is highlighted in red in the first row.

Work prior...	Locked by	Work ID	Work status	Site	Warehouse	Work pool ID	Target license plate ID	Work order ty...
50		USMF-002269	Open	2	24			Sales orders

Work status	Work type	Location	Item number	Product name	Work quan...	Remaining...	Unit	Batch number	User ID
Open	Pick	BULK-007	Z001	Test Product 1	10.00	10.00	ea	LOT84654	
Open	Print		Z001	Test Product 1	10.00	10.00	ea	LOT84654	
Open	Put	BAYDOOR	Z001	Test Product 1	10.00	10.00	ea	LOT84654	

When I complete the picking work on the warehouse mobile application, it now asks me to confirm a specific batch instead of letting me choose whichever one I want.



The screenshot shows the Dynamics 365 F&O Warehouse Management system mobile application interface. The top bar is green and contains 'SO PICK' and 'TASK' tabs. The 'TASK' tab is selected. Below the 'TASK' tab, there is a 'DETAILS' section. The 'DETAILS' section contains the text 'USMF-002269 ✓'. Below the 'DETAILS' section, there is a 'Sales orders : Pick' section. The 'Sales orders : Pick' section contains the text 'CONFIRM BATCH'. Below the 'CONFIRM BATCH' section, there is a large empty rectangular box. At the bottom of the screen, there is a green circle with a white checkmark. The bottom of the screen also displays the text: 'BULK-007, Z001, Qty: 10 ea ; 10 ea, Test Product 1, Batch number: LOT84654'. The 'Batch number: LOT84654' is highlighted in red.

USMF-002269 ✓

Sales orders : Pick

CONFIRM BATCH

BULK-007, Z001, Qty: 10 ea ; 10 ea, Test Product 1, Batch number: LOT84654

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Reservation Hierarchy: Flexible Reservation for Batch tracked items.

Business scenario

In this scenario, a company uses an inventory strategy where finished goods are tracked by batch numbers. This company also uses the WMS workload. Because this workload has well-equipped logic for planning and running warehouse picking and shipping operations for batch-enabled items, most of the finished items are associated with a *Batch-below[location]* inventory reservation hierarchy. The advantage of this type of operational setup is that decisions (which are effectively reservation decisions) about which batches to pick and where to put them in the warehouse are postponed until the warehouse picking operations start. They aren't made when the customer's order is placed.

Although the *Batch-below[location]* reservation hierarchy serves the company's business goals well, many of the company's established customers require the same batch that they previously purchased when they reorder products. Therefore, the company is looking for flexibility in the way that the batch reservation rules are handled, so that, depending on the customers' demand for the same item, the following behaviors occur:

- A batch number can be recorded and reserved when the order is taken by the sales processor, and it can't be changed during warehouse operations and/or taken by other demands. This behavior helps guarantee that the batch number that was ordered is shipped to the customer.
- If the batch number isn't important to the customer, the warehouse operations can determine a batch number during picking work, after sales order registration and reservation have been done.

Allowing reservation of a specific batch on the sales order

To accommodate the desired flexibility in the batch reservation behavior for items that are associated with a *Batch-below[location]* inventory reservation hierarchy, inventory managers must select the Allow reservation on demand order check box for the Batch number level on the Inventory reservation hierarchies page.

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Inventory reservation hierarchies

Name: BatchFlex Description: Batch below flex

Hierarchy

Name	Reservation hierarchy level	Allow reservation on demand order
Site	1	☐
Warehouse	2	☐
Inventory status	3	☐
Location	4	☐
License plate	5	☐
Batch number	6	☒

When the **Batch number** level in the hierarchy is selected, all dimensions above that level and up through the **Location** level will be automatically selected. (By default, all dimensions above the **Location** level are preselected.) This behavior reflects the logic where all dimensions in the range between the batch number and location are also automatically reserved after you reserve a specific batch number on the order line.

Note - The Allow reservation on demand order check box applies only to reservation hierarchy levels that are below the warehouse location dimension.

Batch number and License plate are the only levels in the hierarchy that are open for the flexible reservation policy. In other words, you can't select the Allow reservation on demand order check box for the Location or Serial number level.

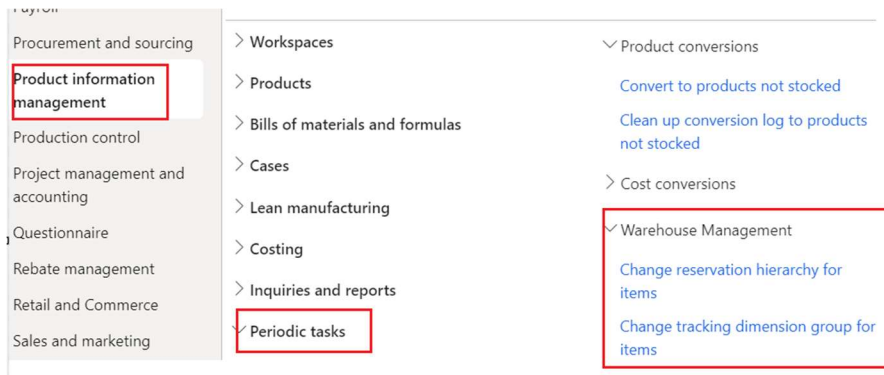
If your reservation hierarchy includes the serial number dimension (which must always be below the Batch number level), and if you've turned on batch-specific reservation for the batch number, the system will continue to handle serial number reservation and picking operations, based on the rules that apply to the Serial-below[location] reservation policy.

At any point, you can allow batch-specific reservation for an existing *Batch-below[location]* reservation hierarchy in your deployment. This change won't affect any reservations and open warehouse work that were created before the change occurred. However, the Allow reservation on demand order check box can't be cleared if inventory transactions of the Reserved ordered, Reserved physical, or Ordered issue type exist for one or more items that are associated with that reservation hierarchy.

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Regardless of whether you've selected the **Allow reservation on demand order** check box, if you don't want to reserve a specific batch number for the item on an order line, default warehouse operations logic that is valid for a *Batch-below[location]* reservation hierarchy will still apply.

How to change the reservation Hierarchy or storage dimension group for item



Change reservation hierarchy for items

Parameters

Name

Records to include

[Filter](#)

ITEMS

Item number

Name – The reservation Hierarchy to be updated to
Item – Input item to be changed

Change tracking dimension group for items

Parameters

New tracking dimension group

New reservation hierarchy

Records to include

[Filter](#)

ITEMS

Item number